Week #8: [Neurovascular Supply of the Distal Upper Extremity]

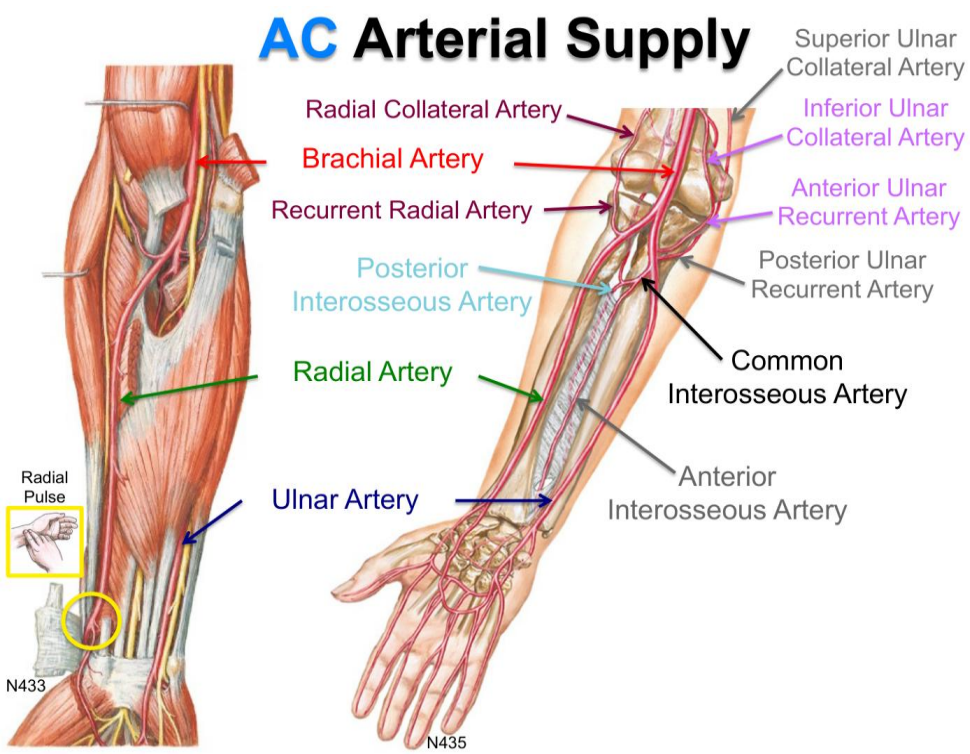
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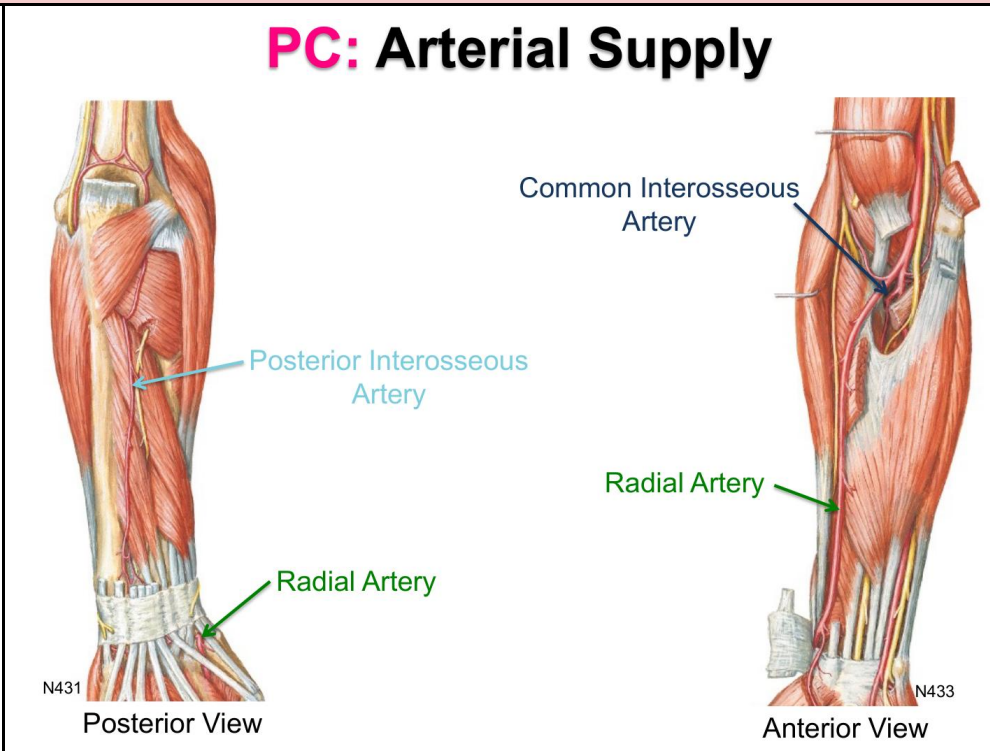
• Introduction to Module

Neurovascular Supply of the forearm

- Vascular Supply of the Forearm (20 min)
- Nerves of the Anterior Forearm (31 min)
- Nerves of the Posterior Forearm (31 min)
- Approach to Nerve Problems at the Elbow (10 min)

Vascular Supply of the Forearm





Vasculature:

a. **Radial Artery** – branch of the brachial artery. Descends through the forearm deep to the brachioradialis. Distally the radial artery wraps around the wrist laterally. The radial pulse can be detected distally between the tendons of flexor carpi radialis and abductor pollicis longus. Proximally the radial artery gives off one branch:
   i. **Recurrent Radial Artery** – will anastomose with the radial collateral artery to contribute to a network of collateral circulation surrounding the elbow.

b. **Ulnar Artery** – branch of the ***brachial artery***. Descends through the forearm deep to pronator teres and enters the wrist lateral to pisiform. Proximally the ulnar artery gives off three branches:
   i. **Anterior Ulnar Recurrent Artery** – will anastomose with the ***inferior ulnar collateral artery*** to contribute to a network of collateral circulation

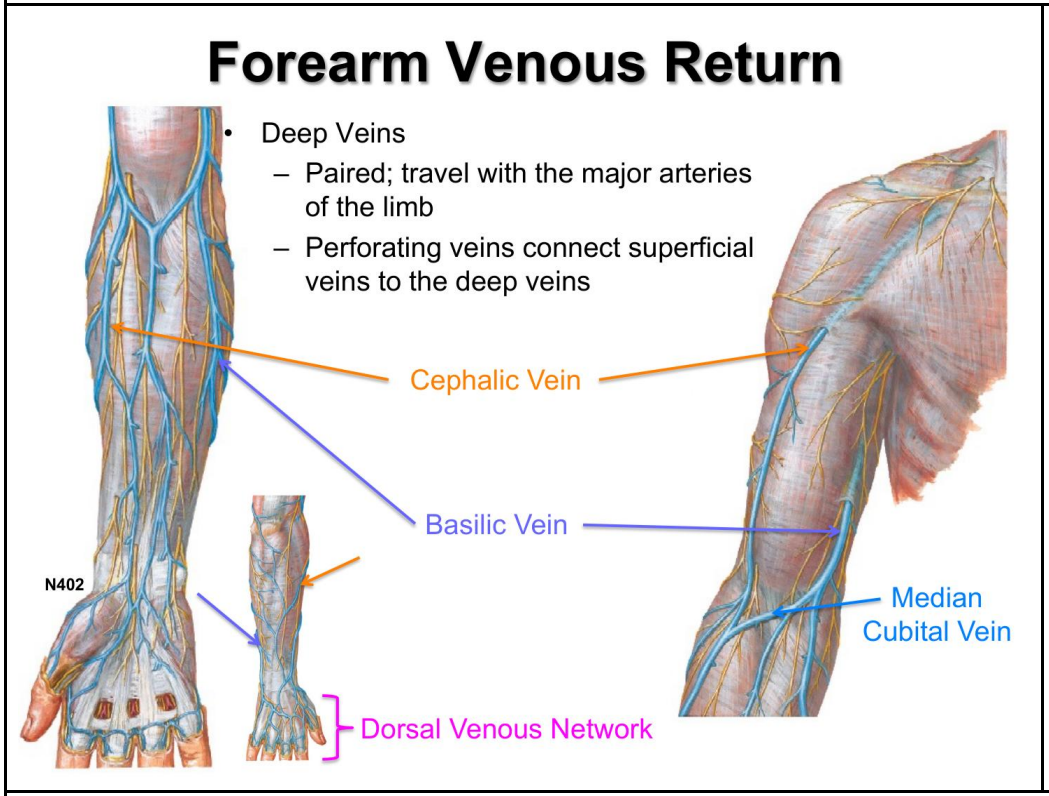
surrounding the elbow.

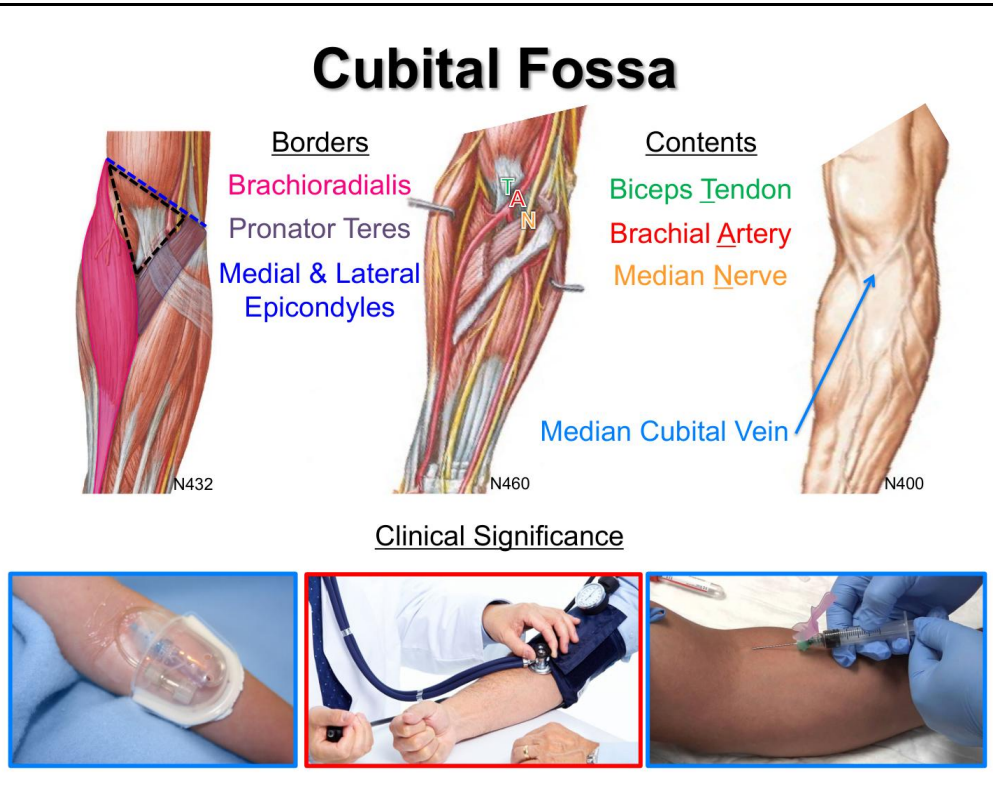
ii. **Posterior Ulnar Recurrent Artery** – will anastomose with the ***superior ulnar collateral artery t***o contribute to a network of collateral circulation surrounding the elbow.

iii. **Common Interosseous Artery** – branches at the distal aspect of the cubital fossa and almost immediately splits into two branches:

1. **Anterior Interosseous Artery** – courses with the ***anterior interosseous nerve*** on the anterior aspect of the interosseous membrane.
2. **Posterior Interosseous Artery** – passes into the posterior compartment of the forearm superior to the interosseous membrane.

Clinical Note: Arterial anastomoses around the elbow provide a functionally and surgically important collateral circulation. If the brachial artery is occluded or needs to be clamped during surgery distal to the origin of the profunda brachii artery the anastomosis will provide the distal arm with sufficient blood supply





II. Anatomical Regions:

a. **Cubital Fossa** – a shallow, triangular depression on the anterior surface of the elbow. Bordered by the brachioradialis, pronator teres, and a line through the medial and lateral epicondyles of the humerus.

b. **Contents:**

i. **Biceps tendon** – easy to palpate; most lateral

ii. **Brachial artery** and veins

iii. **Median nerve**

iv. **Radial nerve** (with deep and superficial branches)

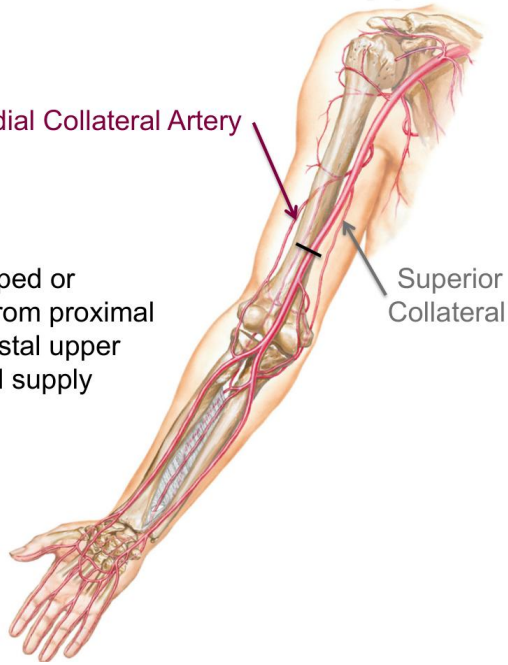
v. Superficial structures:

1. **Median cubital, cephalic** and **basilic veins**
2. **Medial** and **lateral cutaneous nerves** of the **forearm**

B. Anatomical Snuff Box: region on the dorsal hand bounded by the tendons of the APL and EPB anteriorly and posteriorly by EPL. It is visible when the thumb is fully extended. The floor of the snuffbox is comprised of the scaphoid and the trapezium. The radial artery lies in the floor of the box and pulse can be detected here

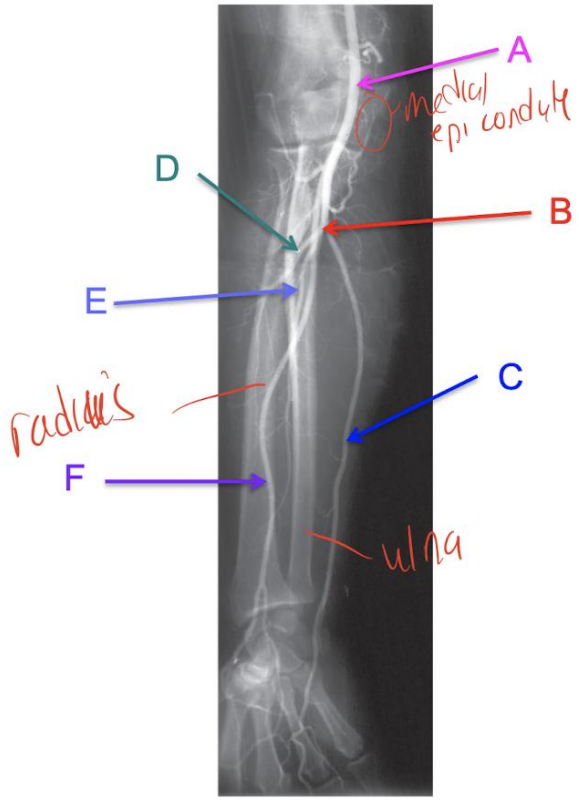
Clinical Note: Scaphoid fracture – the scaphoid is the most commonly fractured bone of the wrist. The scaphoid is small and difficult to immobilize, continued movement of the wrist often results in nonunion of the bone fragments. This may result in avascular necrosis of the fragment if it is separated from its blood supply from the radial artery.

Collateral Circulation of the Upper Limb



- If brachial artery is clamped or occluded blood supply from proximal branches can provide distal upper limb with sufficient blood supply

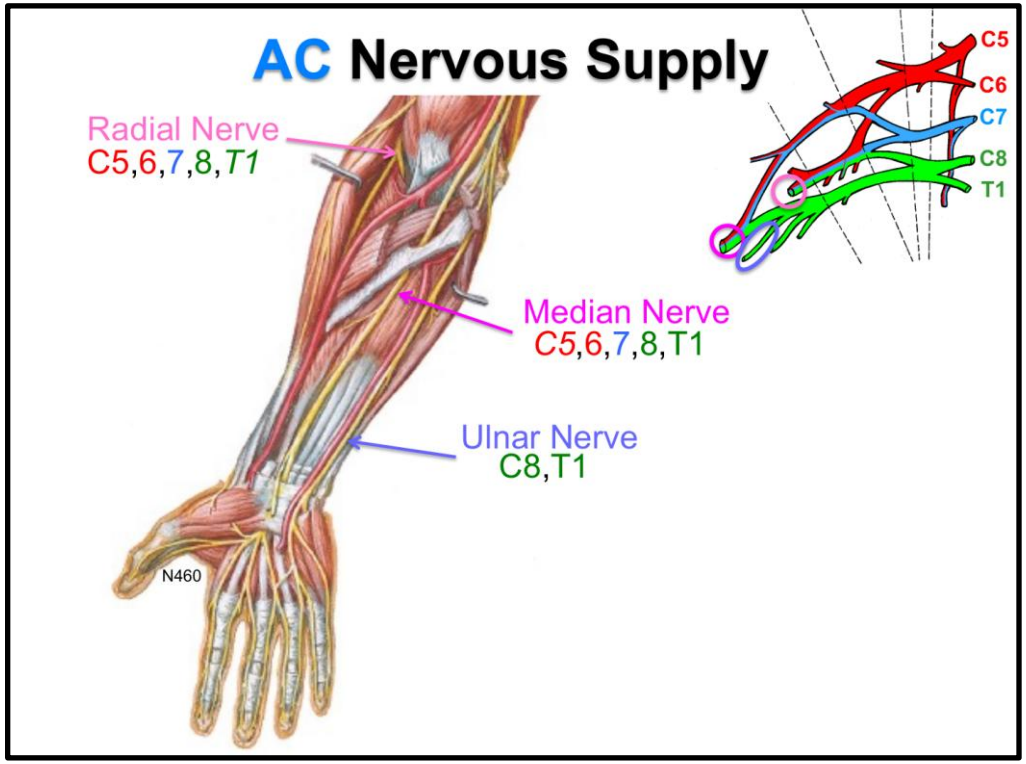
Knowledge Check



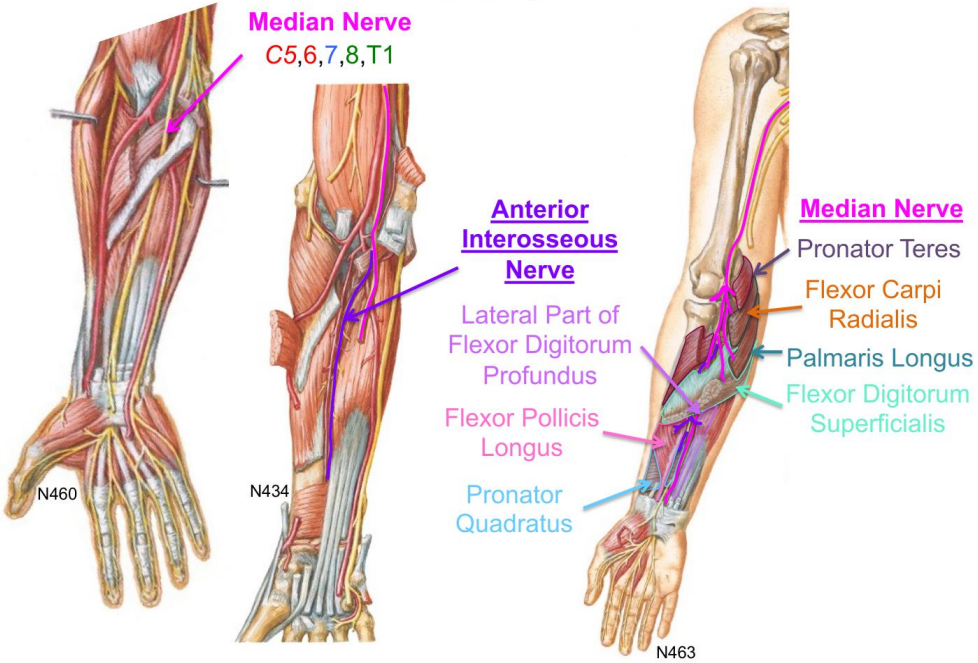
A: brachial artery
B: common interosseous artery
C: ulnar artery
D posterior interosseous artery
E: anterior interosseous artery
F: radial artery

Nerves of the Anterior Forearm

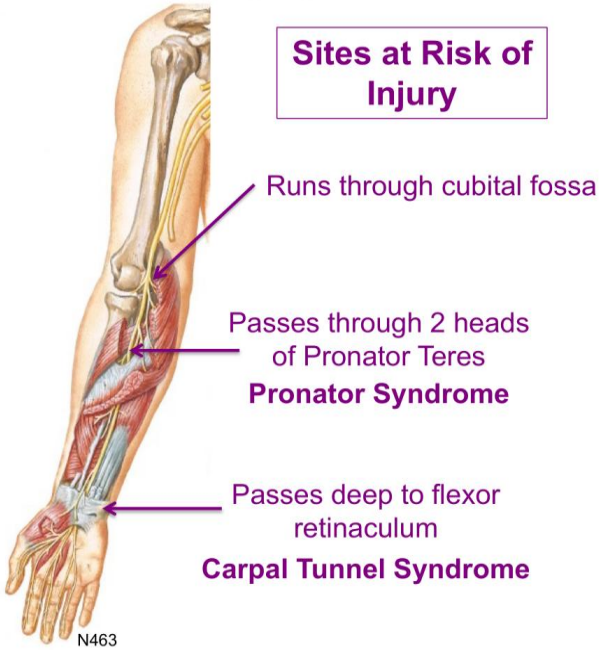
AC Nervous Supply



AC Nervous Supply: Median Nerve



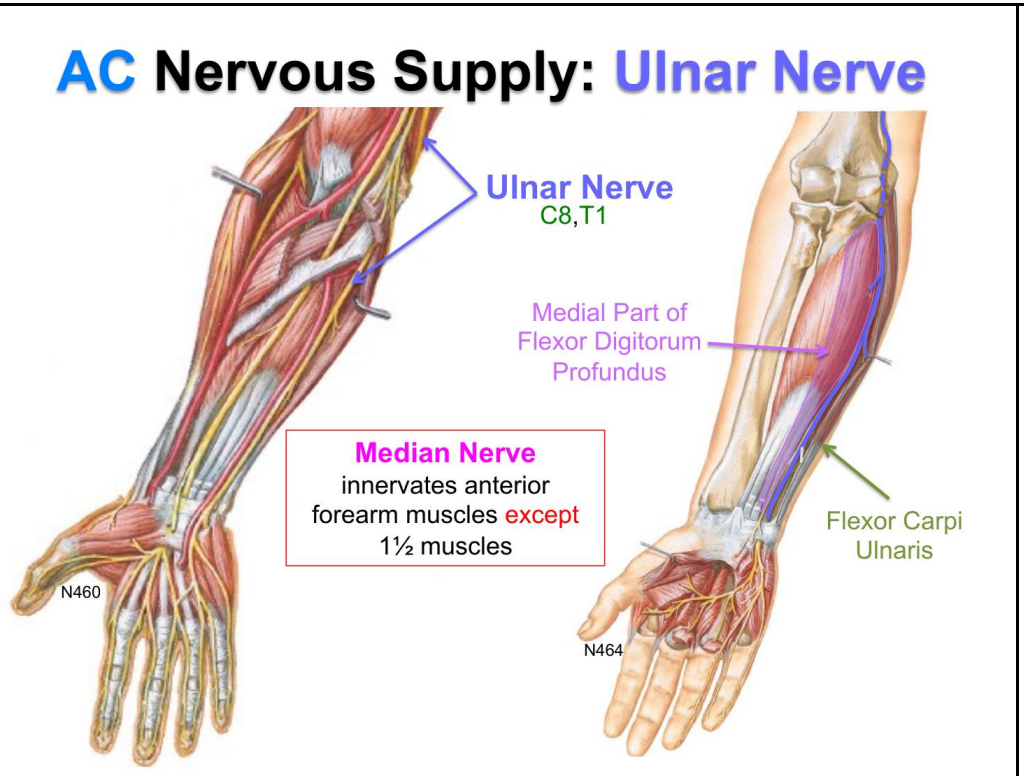
AC Nervous Supply: Median Nerve

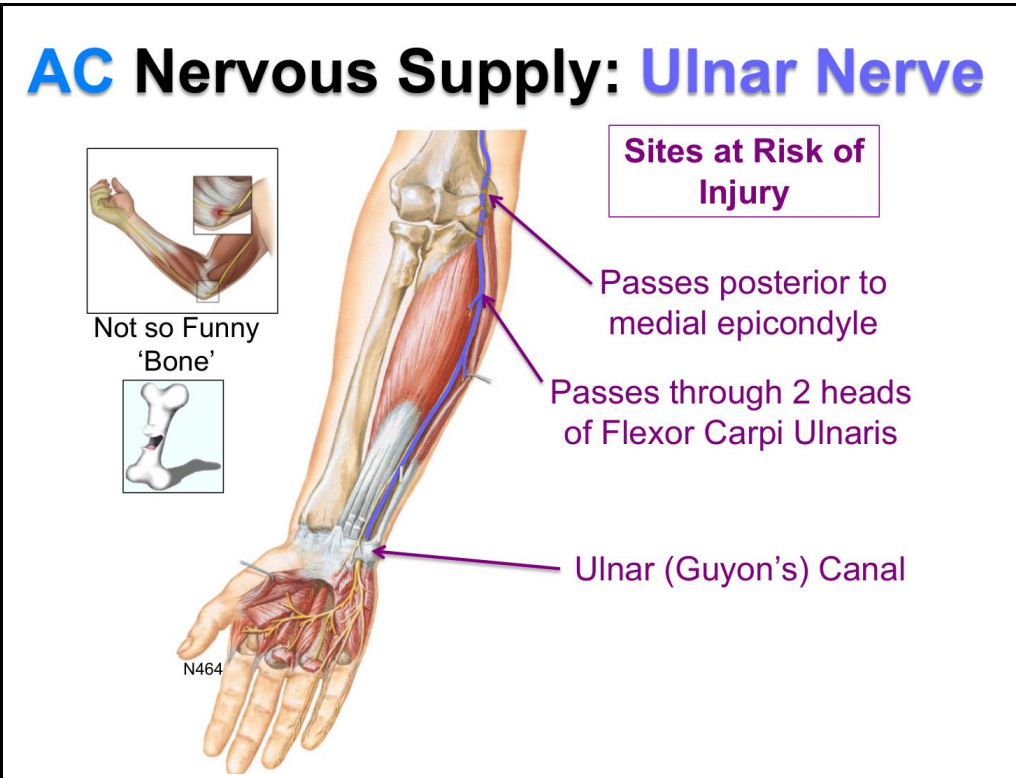


III. Nervous Structures:

a. **Median Nerve** – Descends through the cubital fossa deep to the bicipital aponeurosis and the median cubital vein, medial to the brachial artery. It passes between the heads of the pronator teres then continues distally between the flexor digitorum superficialis and profundus. The median nerve gives rise to ***anterior interosseous nerve*** just distal to the inferior border of the cubital fossa. Median nerve and its branch the anterior interosseous are responsible for the innervation of all muscles in the anterior forearm except flexor carpi ulnaris and the medial half of flexor digitorum profundus.
   i. **Anterior Interosseous Nerve** – branch of the Median Nerve courses through the forearm just anterior to the interosseous membrane with the ***anterior interosseous artery.***
   ii. <u>Sites at risk for injury:</u>
      1. Severed in cubital fossa
      2. Within carpal tunnel (see Carpal Tunnel below)
      3. Compression between the two heads of pronator teres resulting in ***pronator syndrome.***

Clinical Note: **Pronator Syndrome** is a nerve entrapment syndrome caused by compression of the ***median nerve*** between the two heads of pronator teres. It presents with pain and tenderness in the proximal aspect of the anterior forearm and hyperesthesia of the palmar aspects of the lateral 3 ½ digits and adjacent palm. Pronator syndrome is usually a result of muscular hypertrophy from elbow overuse



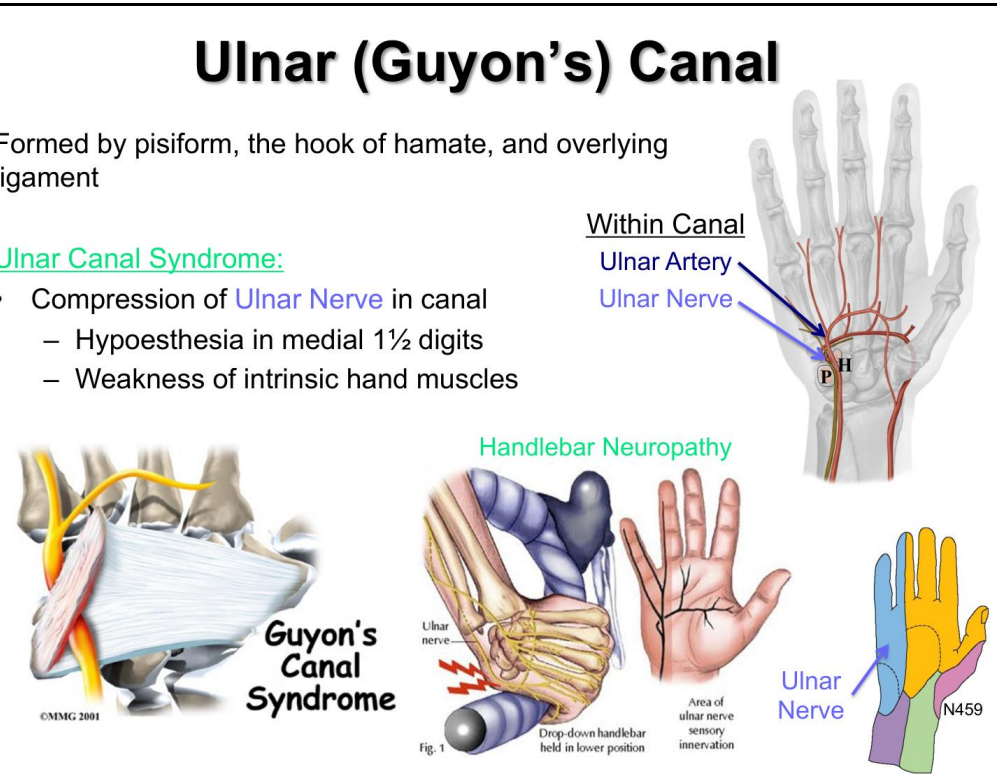


Ulnar Nerve – leaves the arm by wrapping posteriorly around the medial epicondyle. It then courses anteriorly between the heads of the flexor carpi ulnaris and travels through the anterior compartment of the forearm between flexor carpi ulnaris and flexor digitorum profundus.

a. <u>Sites at risk for injury:</u>
   i. Posterior to the medial epicondyle
   ii. Compression within the two heads of flexor carpi ulnaris

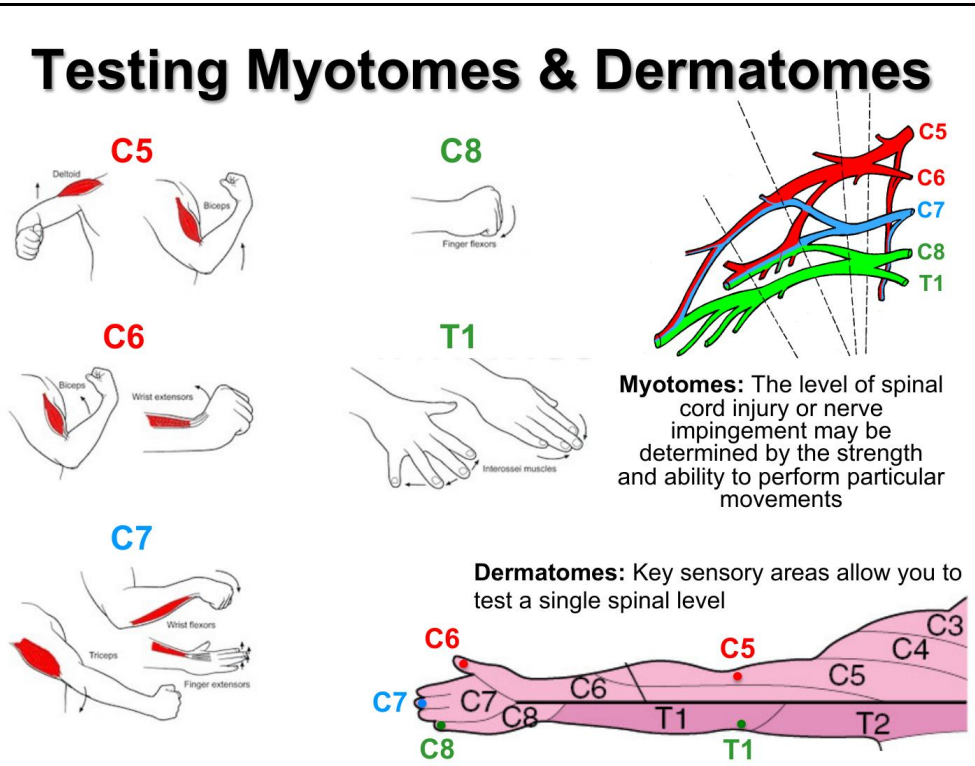
iii. **Ulnar (Guyon's) Canal** (see below)

Clinical Note: As the ulnar nerve passes posterior to the medial epicondyle it is in close approximation to the skin. As a result the nerve can become compressed from external trauma to the elbow resulting in a tingling or pain sensation in the forearm commonly referred to as '**hitting your funny bone'.**



Clinical Note: The **ulnar canal (Guyon's canal)** is formed by the ***pisiform,*** the ***hook of hamate***, and the overlying ligament (pisohamate ligament).

- The ***ulnar artery*** and ***nerve*** travel through this canal. Compression of the ulnar nerve within the ulnar canal may result in hypoesthesia in the medial 1½ digits of the hand and weakness of the intrinsic hand muscles.
- This condition is often known as **handlebar neuropathy**, as riding a bicycle with extended wrists and pressure on the proximal hand (hook of hamate) may result in compression of the ulnar nerve in the ulnar canal



III. Testing Spinal Cord Levels

A. Testing Myotomes:

1. **C5** – Abduction of the arm and flexion of the forearm
2. **C6** – Flexion of the forearm and extension of the wrist
3. **C7** – Extension of the forearm, flexion of the wrist, and extension of the digits
4. **C8** – Flexion of the digits
5. **T1** – Abduction and adduction of the digits

B. Testing Dermatomes:

1. **C5** – Lateral elbow
2. **C6** – Anterior distal 1st digit
3. **C7** – Anterior distal 3rd digit
4. **C8** – Anterior distal 5th digit
5. **T1** – Medial elbow

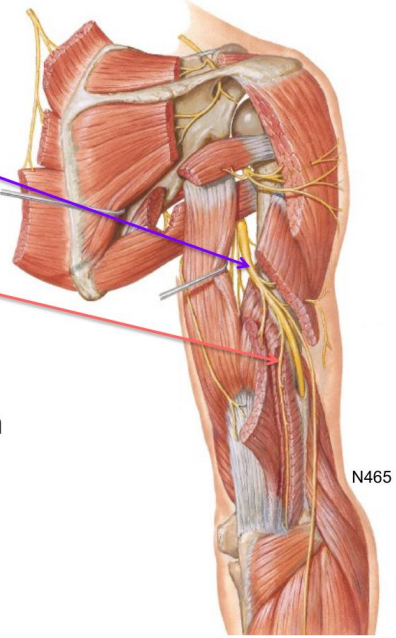
Nerves of the Posterior Forearm

PC: Nerve Supply

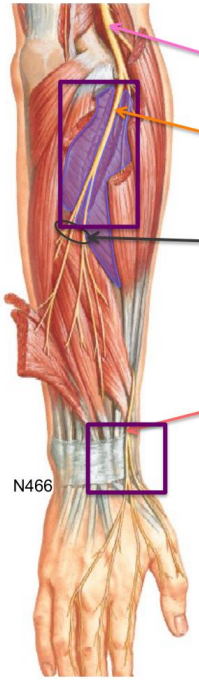
Radial Nerve

Posterior Cutaneous Nerve of Forearm

- Provides sensory innervation to posterior aspect of forearm



PC: Nerve Supply



Posterior View

Radial Nerve

C5 – T1

Deep Branch

- Pierces supinator

Posterior Interosseous Nerve

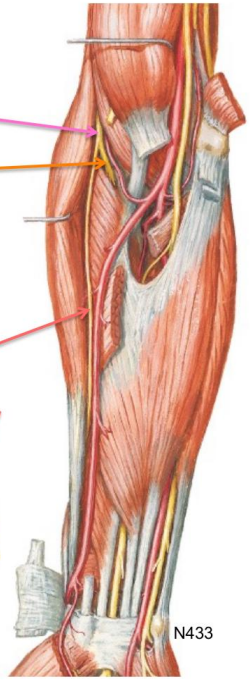
Superficial (Dorsal Sensory) Branch

- SENSORY to dorsum of hand lateral to 4th digit



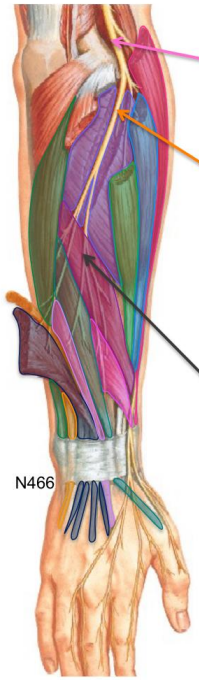
Sites at Risk of Injury

- Pierces supinator
- In anatomical snuff box



Anterior View

PC: Nerve Supply



Posterior View

Radial Nerve

Brachioradialis

Extensor Carpi Radialis Longus

Deep Branch of Radial Nerve

Supinator

Extensor Carpi Radialis Brevis

Extensor Digitorum

Extensor Digiti Minimi

Extensor Carpi Ulnaris

Posterior Interosseous Nerve

Abductor Pollicis Longus

Extensor Pollicis Brevis

Extensor Pollicis Longus

Extensor Indicis

Arm and Forearm Cutaneous Nerve Supply



Axillary Nerve

(Posterior Cord)

Medial Cutaneous Nerve of the Arm

(Medial Cord)

Posterior Cutaneous Nerve of the Arm

(Radial Nerve)

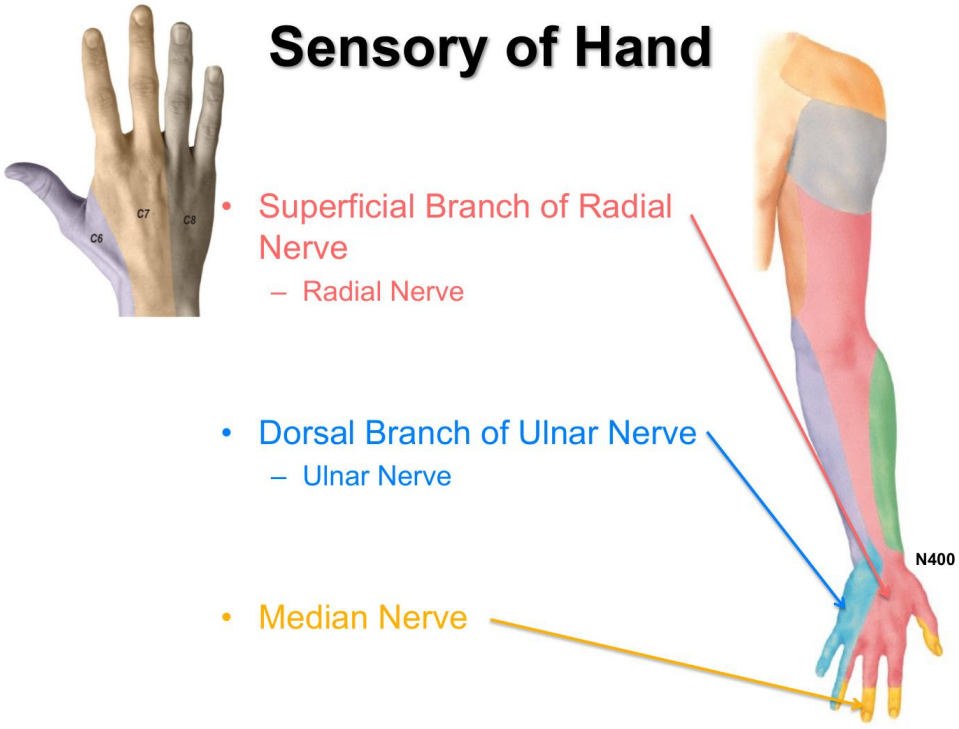
Medial Cutaneous Nerve of the Forearm

(Medial Cord)

Lateral Cutaneous Nerve of the Forearm

(Musculocutaneous Nerve)





Radial Nerve – The radial nerve appears in the cubital fossa between the brachialis and the brachioradialis. Here it divides into two terminal branches:

a. **Superficial (Dorsal Sensory) Branch of Radial Nerve** – is a cutaneous branch that supplies cutaneous innervation to the dorsum of the hand lateral to the 4th digit. This nerve crosses the roof of the anatomical snuffbox.

b. **Deep Branch of Radial Nerve (posterior interosseous nerve)** – it pierces supinator and runs between the superficial and deep extensor muscles. Supplies the majority of the muscles in the posterior compartment of the forearm.

Note: As the tendons of the extensor muscles pass over the dorsal wrist they are enveloped in a **synovial tendinous sheath** to reduce friction from the tendons gliding over the osseous structures.

Approach to Nerve Problems at the Elbow

Neurovascular Supply of the Forearm

Approach to Nerve Problems at the Elbow

Clinical Approach

History	Physical	Management Approach
• Look for patterns!! • OLDCARS • **Acute** (traumatic, inflammatory) vs **chronic** (degenerative, overuse) • Overlay your knowledge of anatomy (which structures are medial, which are lateral, etc)	• LOOK at it • Is it swollen? Are there obvious deformities? • FEEL it • Where does it hurt to touch? Does it feel hot? • MOVE it • Is ROM restricted? • Specific testing maneuvers for specific structures • Don't forget the nerves!	• What (if any) investigations will help you? • If no red flags, apply your pattern • Educate • Non-pharmacologic • Pharmacologic • Interventional • Remember your patient is a person!

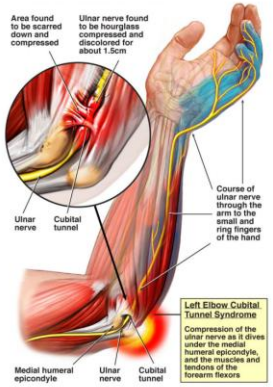
Differential Diagnosis for Elbow Problems

By Anatomic Structure	By Location
• Bone/ligament/joint • Arthritis • Degen, post-traumatic, inflammatory • Subluxation/dislocation • Sprain/tear • UCL • Fracture • Muscle/tendon • Tendinopathies • Medial epicondylosis (golfer's elbow) • Lateral epicondylosis (tennis elbow) • Distal biceps • Distal triceps • Other structures • Olecranon bursitis • Ulnar neuritis/neuropathy	• Anterior • Arthritis • Degenerative, post-traumatic • Inflammatory • Distal bicep tendinopathy • Medial • Medial epicondylosis • Ulnar neuritis/neuropathy • UCL sprain/tear • Lateral • Lateral epicondylosis • Posterior • Olecranon bursitis • Arthritis • Triceps tendinopathy (fracture could be anywhere)

Ulnar Neuropathy at Elbow (UNE)

- May be called cubital tunnel syndrome
  - NB: ulnar nerve can also be compressed at the wrist, though this is less common, and of course, this session is all about the elbow.

- Causes of nerve irritation:
  - Direct trauma – hit funny bone
  - Repeated pressure – eg desk
  - Elbow flexion causes nerve tension – sleep
  - Subluxation of nerve with flexion
  - Ligament hypertrophy?
  - Medial epicondylosis (bulk)?



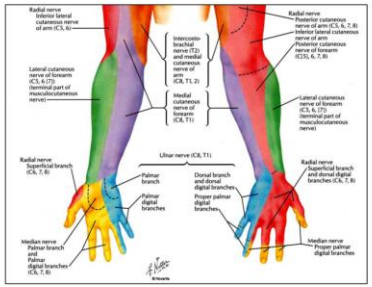
Clinical Features of UNE

- Pain medial elbow
- Numbness & tingling medial forearm into ring and small finger
  - Fingers “fall asleep”
- Positive Tinel Sign at entrapment site
- Check for sensory loss
- Check for motor deficits

Tinel's sign is positive when lightly banging (percussing) over the nerve elicits a sensation of tingling, or 'pins and needles,' in the distribution of the nerve.

Motor and Sensory Features

- Ulnar sensory
  - (NOT MCNFA and NOT C8)



- Ulnar motor
  - May or may not have weakness
  - If they’re weak…
    - Small muscles of hand are usually most affected

- Remember…
  - Tingling/abnormal sensation can come from lots of things
    - Nerve, muscle, tendon, ligament, joint
    - Don’t forget referred – ?cardiac
  - True sensory LOSS is neurogenic

Ulnar Neuropathy at Elbow

- Management (from Approach to Management of ALL THINGS MSK)
  - Rule out red flags – if there is significant or progressive weakness, this might need more aggressive treatment at earlier stage
    - EMG/NCS will be helpful in identifying diagnosis and determining severity
  - Education – critical here!
    - Often there is nerve *irritation* (neuritis) without nerve *damage* (neuropathy)…can distinguish with EMG/NCS
      - Neuritis is totally reversible
      - Mild neuropathy is also totally reversible!
  - Non-pharm
    - Avoid external pressure (don't lean on it!); consider ergonomics!
    - Avoid hyperflexion of elbow (towel splint or elbow pad overnight)
  - Pharm
    - Not much!
  - Intervention/surgery
    - Ulnar neurolysis or ulnar nerve transposition surgery

